Individual Coursework Submission Form

Specialist Masters Programme

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SMM634- Individual Assignment

Modelling Healthcare Expenditure and Utilisation Using MEPS Data

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Abstract

This study investigates the determinants of health-care utilisation and expenditure using cross-sectional data from the Medical Expenditure Panel Survey (MEPS). Two outcomes are analysed: expenditure on doctor visits (dvexpend) and number of doctor visits (dvisit). Based on distributional characteristics, a two-part model is selected for analysing expenditure and a Negative Binomial regression is chosen for utilisation. The results indicate that poorer perceived health, chronic conditions, age, and socio-economic factors significantly increase both health-care utilisation and expenditure. These findings provide policy-relevant evidence illustrating how demand for health services varies across demographic and health-status groups and highlight the importance of need-based allocation of healthcare resources.

Key words: *Negative Binomial Regression, Expenditure on Doctor Visits, Number of Doctor Visits, Two-part Model*

1. Introduction

Health-care systems internationally face increasing pressure from rising utilisation and associated expenditure. The MEPS survey provides rich individual-level data on demographics, health status, socioeconomic characteristics and health-care utilisation, making it well suited for analysing variation in service use and spending.[1]

To contextualize the analysis, Table 1 summarizes key demographic and health characteristics of the MEPS sample. The sample contains slightly more women than men and a notable prevalence of chronic illness, with over a quarter reporting hypertension.

Table 1: *Demographic and health characteristics of the analysis sample.*

Variable	**Mean**	**SD**
Age (years)	40.24	13.66
Female	52.64	–
Male	47.36	–
BMI	28.01	6.41
Log household income	10.37	2.04
General Health: Poor	2.84	–
Mental Health: Poor	1.41	–
Hypertension	25.24	–
Hyperlipidemia	22.65	–

This study aims to:

1. Describe the distributional characteristics of healthcare expenditure and utilisation outcomes.
2. Identify and justify suitable econometric models for expenditure and utilisation.
3. Estimate the effects of demographic and health-status variables.
4. Evaluate model performance and discuss policy implications.

The outcomes studied are:

1. **dvexpend** – individual expenditure on doctor visits
2. **dvisit** – individual number of doctor visits in the year

[1] Agency for Healthcare Research and Quality (AHRQ). 2024. *Medical Expenditure Panel Survey (MEPS)*

These outcomes are modelled separately because expenditure is continuous and highly skewed with many zero values, while utilisation is a discrete count variable with overdispersion.

2. Model Selection for Health-Care Expenditure (dvexpend)

2.1 Descriptive Distribution of Expenditure

Expenditure values demonstrate substantial variation and a large zero mass.

Table 2: *Summary statistics for dvexpend that represents annual spending on doctor visits.*

Statistic	**Value**
N	10,638
Mean	481.07
Median	63
Std. Dev.	1,646.791
Minimum	0
Maximum	59,418
Share of zero values	0.4592969

Expenditure is characterised by:

- **Nearly half zero values**, representing individuals without any doctor visits.
- **Strong positive skew**, with a minority driving high expenditures.
- **Variance > mean**, indicating heteroskedasticity

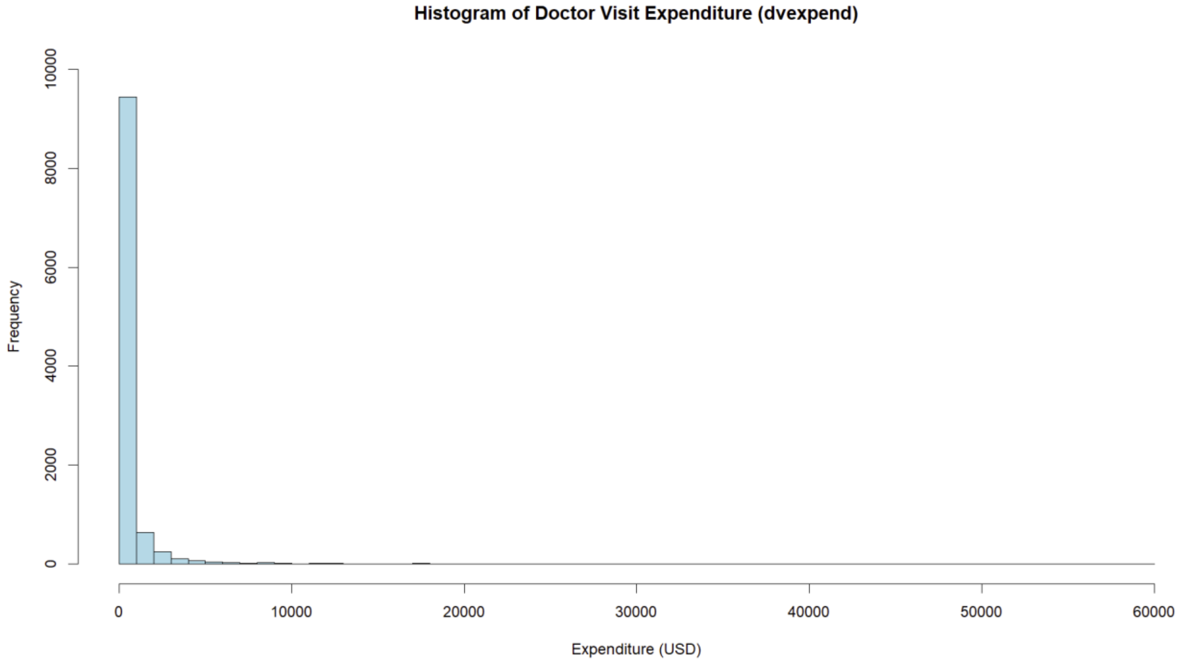

***Figure 1**: Histogram of dvexpend*

The histogram is extremely right-skewed, with almost half of respondents reporting zero spending and a small number incurring very high costs.

2.2 Selection of a Two-Part Model

Two-part models are widely used in health economics to analyze expenditure with a large proportion of zero observations (Manning and Mullahy, 2001)[2]. They separate the decision to seek care from the intensity of spending among users. Due to zero inflation and skewness, a two-part model is used:

Part 1: Probability of any expenditure

Binary dependent variable (as shown in Eq. 1):

$$\boldsymbol{any_{dv_i} = \mathbb{1}(dvexpend_i > 0)} \tag{1}$$

Modelled using Logistic regression (as shown in Eq. 2):

$$\mathbf{Pr}\ (\boldsymbol{any\_dv_i} = \mathbf{1} \mid \boldsymbol{X_i}) = \mathbf{logit^{-1}}(\boldsymbol{X_i \beta}) \tag{2}$$

[2] Manning, W.G. and Mullahy, J., 2001. Estimating healthcare costs with two-part models. *Health Services Research*, 36(4), pp.963–989.

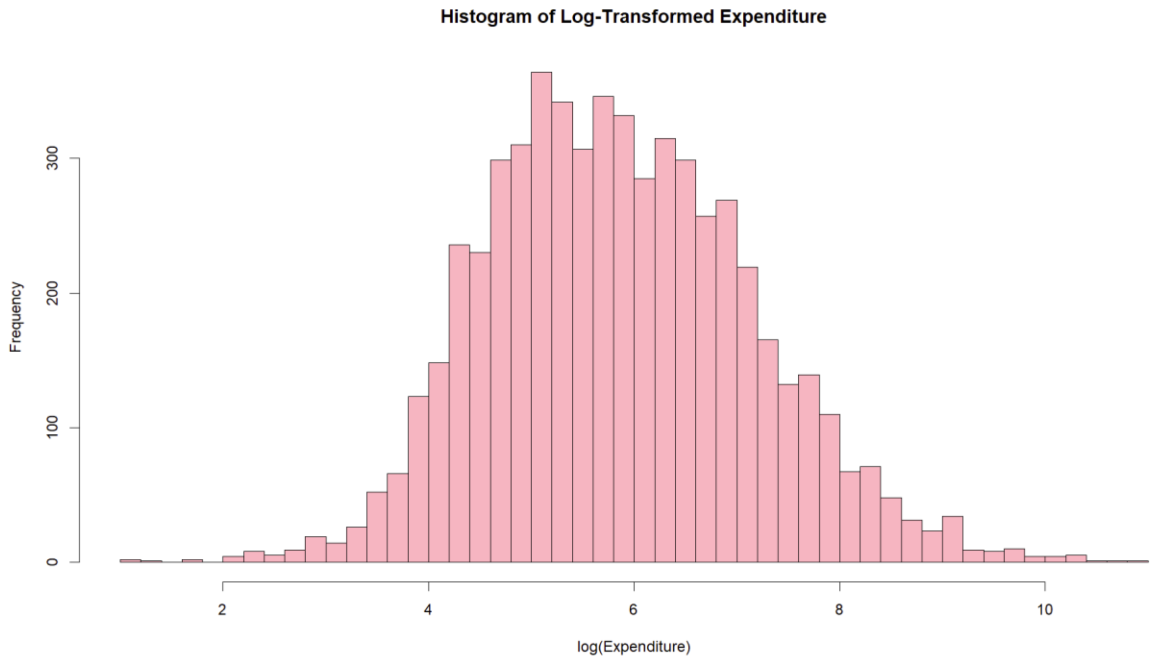

Figure 2: *Histogram of log(dvexpend) for positive expenditure*

This histogram produced to assess the distribution of positive spending amounts. The log transformation greatly reduces right skewness and produces a more symmetric distribution, supporting the use of a Gamma model with a log link for the second part of the two-part expenditure model.

Part 2: Level of expenditure conditional on use

For positive dvexpend, the conditional distribution is positive and skewed. A Gamma GLM with log link is appropriate (as shown in Eq. 3):

$$\boldsymbol{E(dvexpend_i \mid dvexpend_i > 0, X_i) = \exp\,(X_i\gamma)} \tag{3}$$

- The log link ensures **positive fitted values** and allows **multiplicative** interpretations of coefficients.

This two-part model explicitly separates:

- the **decision to use** doctor services, and
- the **level of expenditure** among users,

which is substantively plausible.

2.3 Covariates

To ensure comparability, both parts use the same predictors:

Variable	**Description**
general, mental (factor)	self-reported physical & mental health (1=excellent to 5=poor)
age (numeric), gender (binary indicators)	demographics
income (numeric)	log household income
education (numeric)	years of education
ethnicity, region (factor)	categorical controls
bmi (numeric)	body mass index
hypertension, hyperlipidemia (binary indicators)	chronic conditions

This choice is motivated by both **economic theory** and the **data description**.

3. Model Selection for Health-care Utilisation (dvisit)

3.1 Distributional Characteristics

The outcome dvisit is a non-negative integer count variable.

Table 3: *Summary statistics for (dvisit)* the **number of doctor visits** over the year

Statistic	**Value**
N	10,638
Mean	2.1316
Median	1
Variance	12.14184
Minimum	0
Maximum	29
Share of zero values	0.4592969

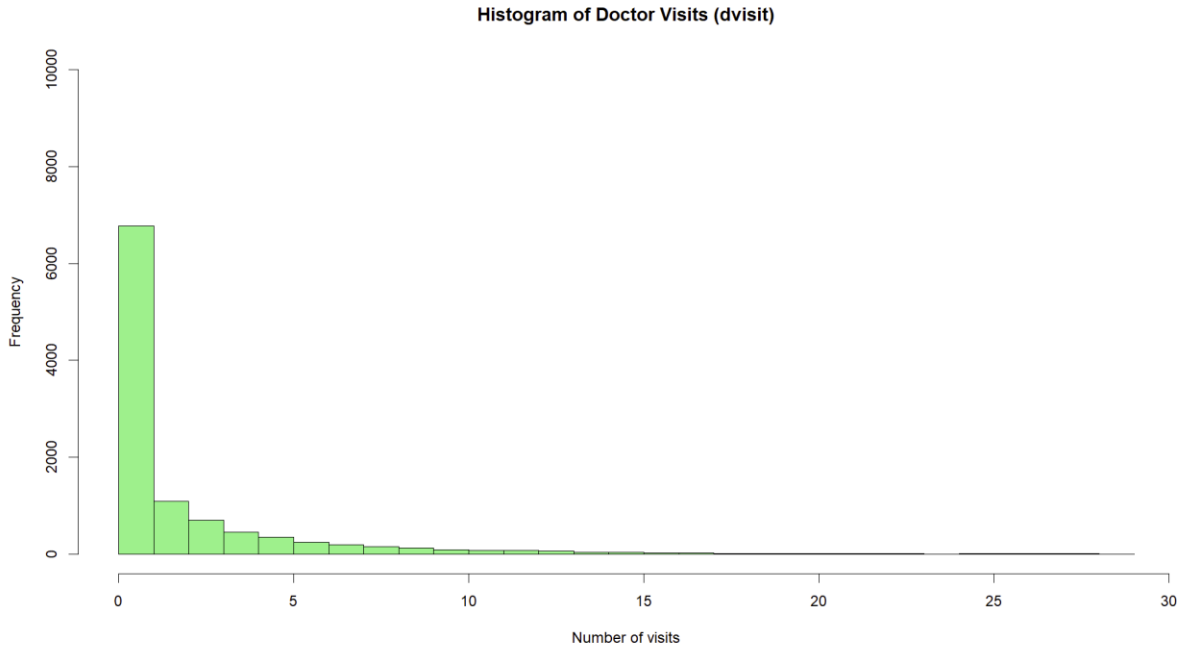

Figure 3: *Histogram of dvisit*

The plot shows heavy right skew and a large spike at zero, indicating that many individuals made no visits while relatively few had high utilization. The **Variance > mean**, confirming overdispersion this indicates that visit behaviour varies more across individuals than the Poisson model assumes, resulting in greater spread and heavier tails in the distribution. and supporting the choice of a NB model over Poisson assumption **Var = Mean**[3].

3.2 Poisson vs Negative Binomial

As a starting point, we can fit a Poisson regression (as shown in Eq. 4):

$$\mathbb{E}(\mathbf{dvisit}_i \mid \boldsymbol{X}_i) = \mathbf{exp}\,(\boldsymbol{X}_i'\boldsymbol{\beta}), \mathbf{dvisit}_i \sim \mathbf{Poisson}(\cdot) \tag{4}$$

and check for overdispersion via:

- residual deviance vs degrees of freedom, and/or
- a formal dispersion test.

The observed variance being much larger than the mean, and the presence of many zeros, strongly suggests **overdispersion** in the Poisson.

Test	**Statistic**	**p-value**
Dispersion test	9.72	< 0.001

→ **Reject Poisson** due to overdispersion.

[3] Cameron, A.C. and Trivedi, P.K., 2013. *Regression Analysis of Count Data*. Cambridge: Cambridge University Press.

To account for overdispersion, we move to a **(NB) regression**[4] (as shown in Eq. 5 and 6):

$$\mathbb{E}(\textbf{dvisit}_i \mid \boldsymbol{X}_i) = \textbf{exp}\,(\boldsymbol{X}_i'\boldsymbol{\beta}), \textbf{Var}(\textbf{dvisit}_i \mid \boldsymbol{X}_i) = \mathbb{E}(\textbf{dvisit}_i \mid \boldsymbol{X}_i)[\mathbf{1} + \boldsymbol{\alpha}\mathbb{E}(\textbf{dvisit}_i \mid \boldsymbol{X}_i) \quad (5)$$

$$\boldsymbol{Var(Y) = E(Y) + \alpha E(Y)^2} \quad (6)$$

where $\alpha > 0$is an overdispersion parameter estimated from the data.

Model comparison using AIC:

Model	**AIC**
Poisson	51956.65
NegBin	37795.85

→ The NB AIC is substantially lower, confirming that NB is the preferred model due to overdispersion.

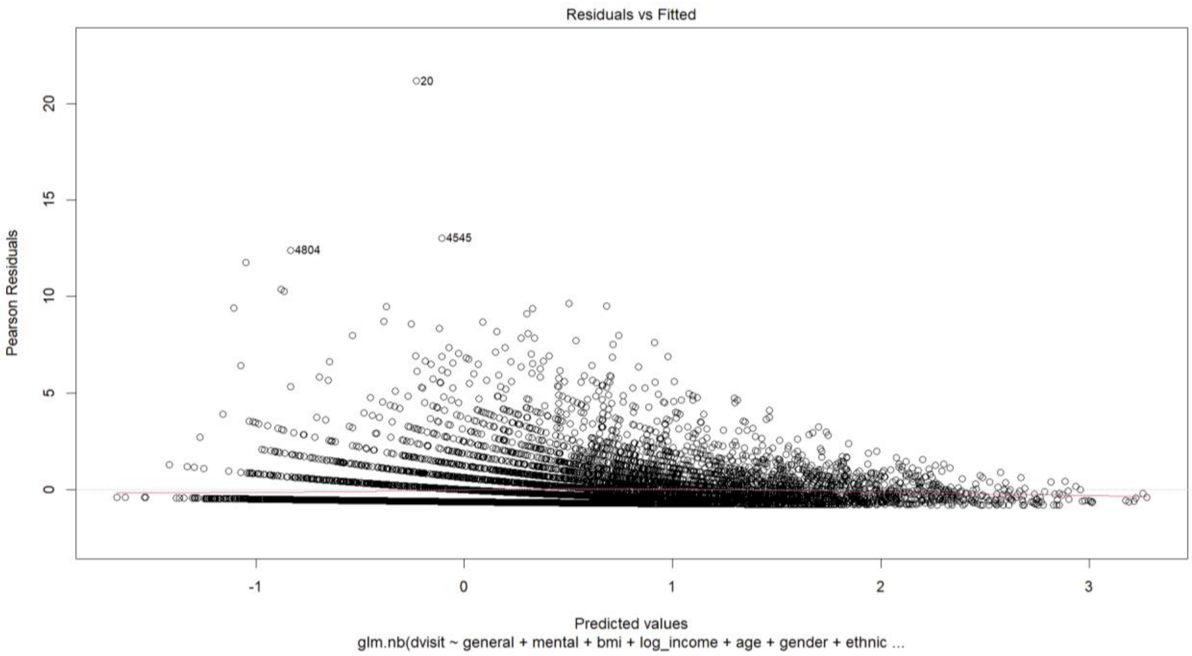

Figure 4a: *Residual vs fitted values*

The NB model was examined to assess model adequacy. The absence of a clear pattern in residual distribution suggests that the NB model appropriately handles variance structure in the utilization data.

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[4] Hilbe, J.M., 2011. *Negative Binomial Regression*. Cambridge: Cambridge University Press.

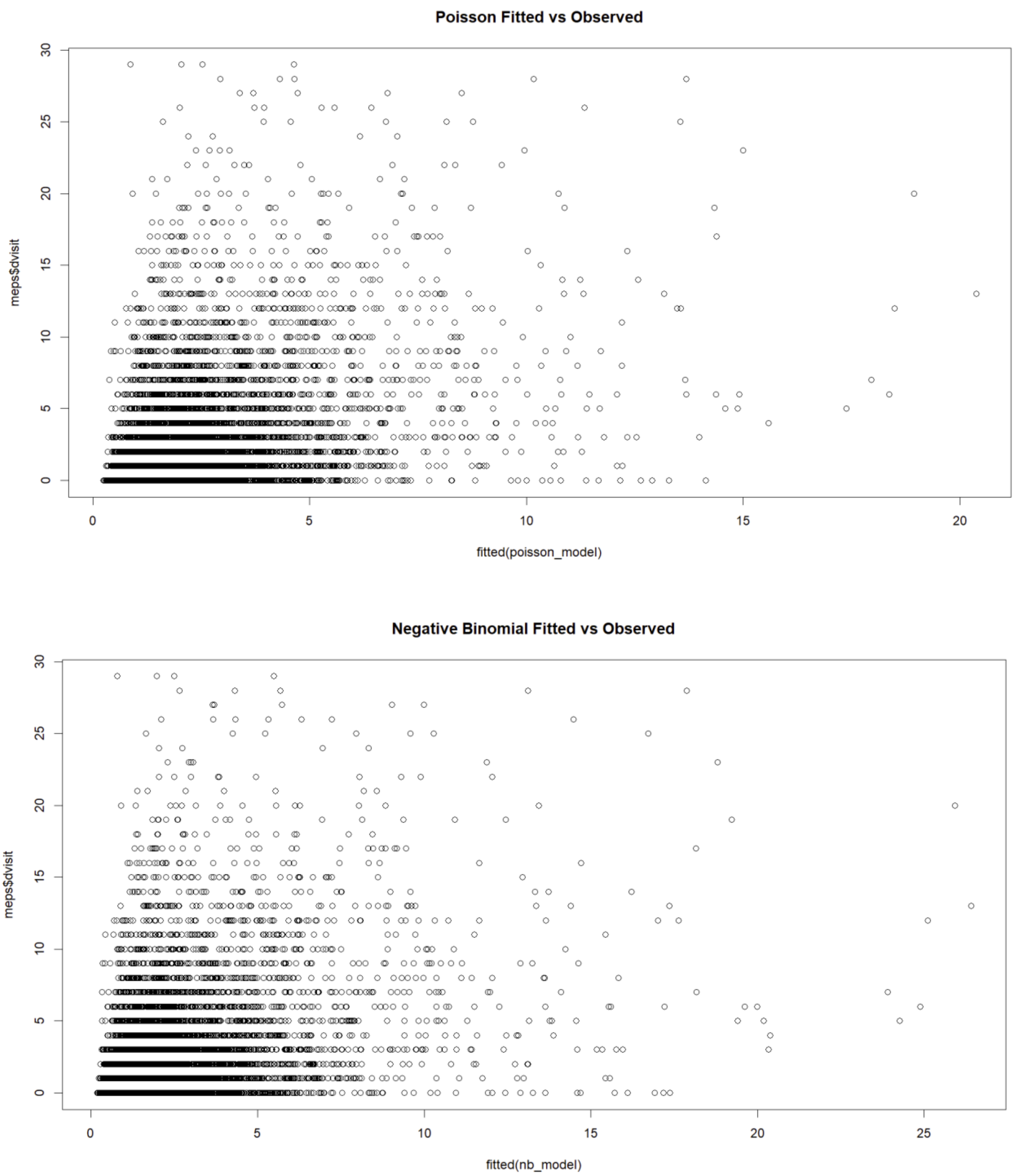

Figure 4b: *Fitted vs observed (Poisson vs NB) comparison*

The plots shows that the Poisson model underestimates high visit counts due to overdispersion, while the NB model aligns much more closely with actual values, demonstrating its superior fit.

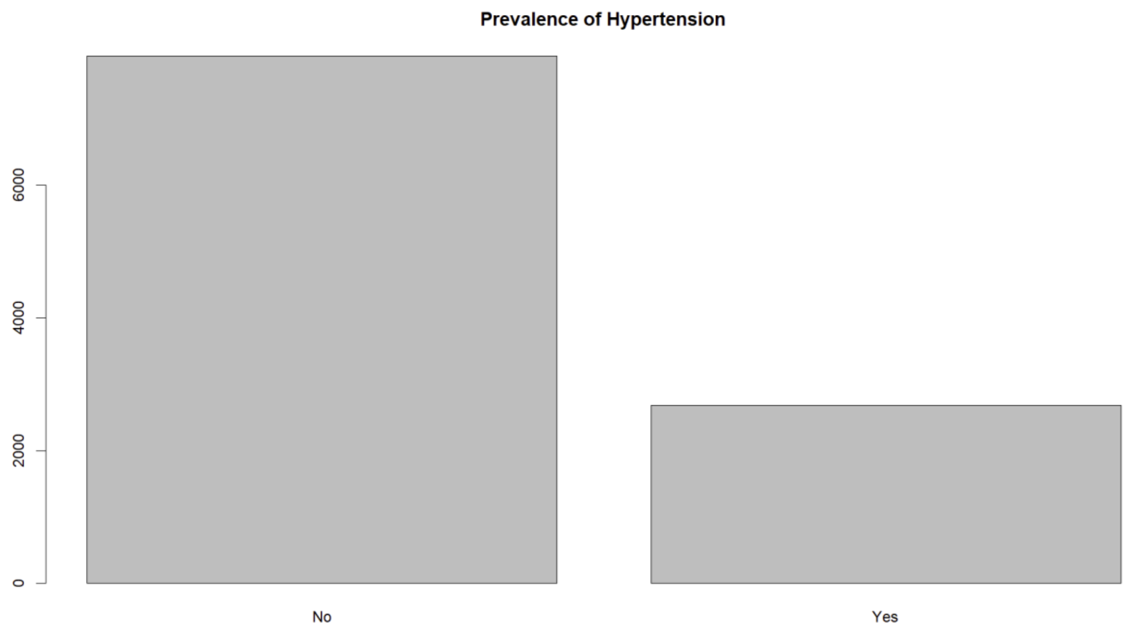

Figure 4c: *Hypertension prevalence bar chart*

The chart shows a large proportion of individuals with a chronic condition, reinforcing its importance as an explanatory variable in modelling health-care expenditure and utilization.

Overall, the diagnostic plots support the chosen modelling strategies: the log-transformed expenditure histogram confirms the suitability of the Gamma model and the fitted-versus-observed comparisons show that the NB specification fits utilization data much better than the Poisson. The hypertension prevalence chart further supports treating chronic conditions as key predictors.

4. Empirical Results

4.1 Two-Part Expenditure Model (dvexpend)

Part 1: Logistic regression for any spending

Table 4: *Factors affecting likelihood of any expenditure*

Variable	Odds Ratio	Interpretation
General health = Poor	3.92	~4x more likely to have any expenditure
Mental health = Poor	1.40	More likely to seek care
Hypertension	1.81	Higher likelihood of use
Hyperlipidemia	2.21	Higher likelihood of use
Gender (Male)	0.397	Men less likely than women
Age	1.02	Each year increases odds

Log Income	1.06	Higher income increases access

Poor physical health, chronic illness, age, and higher income increase the likelihood of accessing health-care services, while male respondents are substantially less likely than females to incur any doctor visit expenditure.

Part 2: Gamma model

Table 5: *Determinants of expenditure among users*

Variable	Exp(β)	Interpretation
General health = Poor	3.13	~3x higher expenditure among users
Hypertension	1.13	27% higher spending
Age	1.01	Small but significant growing effect
Gender (Male)	0.804	Men spend 22% less than women

Among users, poorer general health and chronic disease lead to substantially higher spending, and men consistently incur lower expenditure than women.

Importantly, the results from the logistic and Gamma components are consistent with each other and this consistency strengthens confidence in the two-part structure, as the same underlying drivers shape both the decision to use care and the intensity of spending among users.
Overall, the two-part model suggests that poor general health, older age, female gender, higher income and chronic conditions are key drivers of higher doctor-visit spending.

4.2 Negative Binomial Model (dvisit)

Table 6: *Determinants of doctor visit frequency*

Variable	IRR (Incidence Rate Ratio)	Interpretation
General health = Poor	3.47	3.4x more visits
Mental health = Poor	1.89	Higher use
Hypertension	1.36	More frequent care
Age	1.01	More visits with age
Gender (Male)	0.508	Men make 48% fewer visits

Poor general and mental health, chronic illness, and age increase utilization volume, while men demonstrate significantly lower levels of utilization.

Together, these findings indicate that the **same set of predictors** that drive expenditure also drive utilisation: worse health, chronic conditions, higher age and better socio-economic status all increase the **frequency** of doctor visits.

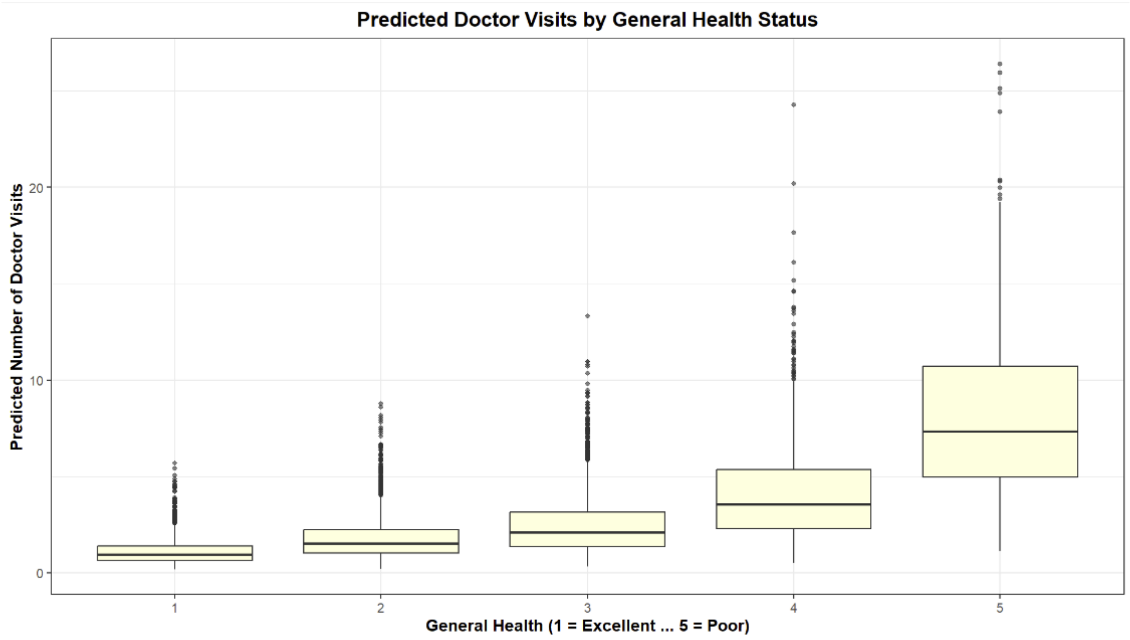

Figure 5: *Predicted visits by health status*

The magnitude of predicted utilization increases sharply with worsening health. This effect size is large in policy terms: individuals with poor health utilize approximately five times more services than those reporting excellent health, reflecting the disproportionate burden on healthcare resources.

5. Analysis Evaluation and Connections

5.1 Strengths

The modelling approach offers several strengths. The two-part model appropriately handles zero expenditure values by separating the decision to seek care from the level of spending, while the Gamma-log specification effectively models positive, skewed costs. The NB model addresses overdispersion in utilization counts and provides more reliable estimates than the Poisson alternative and using consistent covariates across models enables direct comparison.

5.2 Limitations

The use of cross-sectional observational data limits causal inference and leaves results susceptible to unobserved confounding, such as differences in insurance coverage or health-seeking behaviour. The independence assumption in the two-part model may also be restrictive. Model-specific limitations include modest explanatory power in the GM and possible remaining excess zeros in the NB model, suggesting that zero-inflated models could be explored in future work.

5.3 Connections between Models

Both models identify similar determinants: poorer health, chronic illness, older age and higher income all increase utilization and expenditure, while men consistently use and spend less. These parallel findings reinforce the robustness of the overall results and highlight the importance of health need and socioeconomic factors in shaping health-care demand.

6. Conclusion

This study examined determinants of health-care utilization and expenditure using MEPS data and applied models suited to the distributional characteristics of each outcome. The two-part model effectively captured zero-inflated, skewed expenditure data, while the NB model provided a strong fit for overdispersed visit counts. Across both outcomes, poorer health, chronic conditions, age and income emerged as key drivers of demand, with men consistently using and spending less. Diagnostic checks further confirmed that the chosen model specifications were appropriate. Overall model adequacy is further supported by residual plots and fitted-versus-observed diagnostics, which show that the Gamma and NB models capture the key distributional features of the expenditure and utilization data..

References

Cameron, A. Colin, and Pravin K. Trivedi. *Regression Analysis of Count Data*. *Cambridge University Press*, 2nd ed., Cambridge, Cambridge University Press, 2013, www.cambridge.org/core/books/regression-analysis-of-count-data/2AB83B406C5798030F7C91ECC99B1BE4.

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Jones, Am, et al. *Applied Health Economics*. *Routledge EBooks*, Informa, 1 Dec. 2013, www.taylorfrancis.com/books/mono/10.4324/9780203102411/applied-health-economics-andrew-jones-nigel-rice-teresa-bago-uva-silvia-balia. Accessed 21 Apr. 2023.

Manning, Willard G, and John Mullahy. "Estimating Log Models: To Transform or Not to Transform?" *Journal of Health Economics*, vol. 20, no. 4, July 2001, pp. 461–494, https://doi.org/10.1016/s0167-6296(01)00086-8. Accessed 19 Dec. 2022.

"Medical Expenditure Panel Survey Home." *Meps.ahrq.gov*, https://meps.ahrq.gov/mepsweb/.

Appendix

Full R Code For Analysis, Tables & Figures - Rmarkdown

```
# Load required packages
library(MASS)
library(AER)

## Loading required package: car

## Loading required package: carData

## Loading required package: lmtest

## Loading required package: zoo

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric

## Loading required package: sandwich

## Loading required package: survival

library(broom)
library(dplyr)

##
## Attaching package: 'dplyr'

## The following object is masked from 'package:car':
##
##     recode

## The following object is masked from 'package:MASS':
##
##     select

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union

library(ggplot2)


meps <- read.table("meps.txt", header = TRUE)

# Convert categorical variables and create derived variables
meps$general     <- factor(meps$general)
meps$mental      <- factor(meps$mental)
```

```
meps$ethnicity  <- factor(meps$ethnicity)
meps$region     <- factor(meps$region)
meps$gender     <- factor(meps$gender, levels = c(0,1), labels = c("Female
","Male"))
meps$log_income <- log(meps$income + 1)
meps$any_dv     <- as.numeric(meps$dvexpend > 0)   # Indicator for positiv
e expenditure


#TABLE 1: Demographic and health characteristics of the analysis sample
library(dplyr)

demographic_table <- meps %>%
  summarise(
    Mean_Age = mean(age, na.rm = TRUE),
    SD_Age = sd(age, na.rm = TRUE),
    Female = mean(gender == "Female") * 100,
    Male = mean(gender == "Male") * 100,
    Poor_General_Health = mean(general == 5) * 100,
    Poor_Mental_Health = mean(mental == 5) * 100,
    Hypertension = mean(hypertension == 1) * 100,
    Hyperlipidemia = mean(hyperlipidemia == 1) * 100,
    Mean_BMI = mean(bmi, na.rm = TRUE),
    SD_BMI = sd(bmi, na.rm = TRUE),
    Mean_Log_Income = mean(log_income, na.rm = TRUE),
    SD_Log_Income = sd(log_income, na.rm = TRUE)
  )

demographic_table

##   Mean_Age   SD_Age   Female     Male Poor_General_Health Poor_Mental_H
ealth
## 1 40.24788 13.65539 52.64147 47.35853            2.838879           1.4
10039
##   Hypertension Hyperlipidemia Mean_BMI   SD_BMI Mean_Log_Income SD_Log_
Income
## 1     25.23971       22.64523 28.02946 6.411395        10.36539      2.
039794

# TABLE 2: Summary statistics for dvexpend

n <- nrow(meps)
mean_dvexpend   <- mean(meps$dvexpend, na.rm = TRUE)
median_dvexpend <- median(meps$dvexpend, na.rm = TRUE)
sd_dvexpend     <- sd(meps$dvexpend, na.rm = TRUE)
min_dvexpend    <- min(meps$dvexpend, na.rm = TRUE)
max_dvexpend    <- max(meps$dvexpend, na.rm = TRUE)
zero_share_dvexpend <- mean(meps$dvexpend == 0)

cat("Summary of dvexpend:\n",
    "N =", n, "\n",
    "Mean =", mean_dvexpend, "\n",
    "Median =", median_dvexpend, "\n",
    "Standard deviation =", sd_dvexpend, "\n",
```

```
    "Min =", min_dvexpend, "\n",
    "Max =", max_dvexpend, "\n",
    "Share zero =", zero_share_dvexpend, "\n\n")

## Summary of dvexpend:
##  N = 10638
##  Mean = 481.07
##  Median = 63
##  Standard deviation = 1646.791
##  Min = 0
##  Max = 59418
##  Share zero = 0.4592969

# TABLE 3: Summary statistics for dvisit


mean_dvisit   <- mean(meps$dvisit, na.rm = TRUE)
median_dvisit <- median(meps$dvisit, na.rm = TRUE)
var_dvisit    <- var(meps$dvisit, na.rm = TRUE)
min_dvisit    <- min(meps$dvisit, na.rm = TRUE)
max_dvisit    <- max(meps$dvisit, na.rm = TRUE)
zero_share_dvisit <- mean(meps$dvisit == 0)

summary_dvisit <- data.frame(
  Statistic = c("N", "Mean", "Median", "Variance", "Minimum", "Maximum", "
Share of zeros"),
  Value     = c(n, mean_dvisit, median_dvisit, var_dvisit,
                min_dvisit, max_dvisit, zero_share_dvisit)
)

cat("Summary of dvisit:\n",
    "N =", n, "\n",
    "Mean =", mean_dvisit, "\n",
    "Median =", median_dvisit, "\n",
    "Variance =", var_dvisit, "\n",
    "Min =", min_dvisit, "\n",
    "Max =", max_dvisit, "\n",
    "Share zero =", zero_share_dvisit, "\n\n")

## Summary of dvisit:
##  N = 10638
##  Mean = 2.131604
##  Median = 1
##  Variance = 13.14184
##  Min = 0
##  Max = 29
##  Share zero = 0.4592969

summary_dvisit

##        Statistic        Value
## 1              N 1.063800e+04
## 2           Mean 2.131604e+00
## 3         Median 1.000000e+00
## 4       Variance 1.314184e+01
```

```
## 5        Minimum 0.000000e+00
## 6        Maximum 2.900000e+01
## 7 Share of zeros 4.592969e-01

# MODEL SELECTION FOR UTILISATION (dvisit)

# Poisson model
poisson_model <- glm(
  dvisit ~ general + mental + bmi + log_income + age + gender +
    ethnicity + education + region + hypertension + hyperlipidemia,
  family = poisson(link = "log"),
  data = meps
)

summary(poisson_model)

##
## Call:
## glm(formula = dvisit ~ general + mental + bmi + log_income +
##     age + gender + ethnicity + education + region + hypertension +
##     hyperlipidemia, family = poisson(link = "log"), data = meps)
##
## Coefficients:
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -1.4121786  0.0611438 -23.096  < 2e-16 ***
## general2        0.2776220  0.0247251  11.228  < 2e-16 ***
## general3        0.5315867  0.0257142  20.673  < 2e-16 ***
## general4        0.7696846  0.0298924  25.748  < 2e-16 ***
## general5        1.1332446  0.0362569  31.256  < 2e-16 ***
## mental2         0.0217763  0.0207484   1.050   0.2939
## mental3         0.0310412  0.0218408   1.421   0.1552
## mental4         0.3385724  0.0280047  12.090  < 2e-16 ***
## mental5         0.5299033  0.0404335  13.106  < 2e-16 ***
## bmi             0.0092018  0.0009952   9.246  < 2e-16 ***
## log_income      0.0004682  0.0034157   0.137   0.8910
## age             0.0138124  0.0005917  23.345  < 2e-16 ***
## genderMale     -0.5440187  0.0141770 -38.373  < 2e-16 ***
## ethnicity2     -0.1700478  0.0177404  -9.585  < 2e-16 ***
## ethnicity3     -0.1398903  0.0738866  -1.893   0.0583 .
## ethnicity4     -0.1904544  0.0267522  -7.119 1.09e-12 ***
## education       0.0794679  0.0025750  30.861  < 2e-16 ***
## region2         0.0042223  0.0212637   0.199   0.8426
## region3        -0.1809594  0.0196388  -9.214  < 2e-16 ***
## region4        -0.1218735  0.0212103  -5.746 9.14e-09 ***
## hypertension    0.2339695  0.0163826  14.282  < 2e-16 ***
## hyperlipidemia  0.3176492  0.0160222  19.826  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 46449  on 10637  degrees of freedom
## Residual deviance: 35297  on 10616  degrees of freedom
## AIC: 51957
```

```
##
## Number of Fisher Scoring iterations: 6
```

```
dispersiontest(poisson_model)   # Overdispersion test
```

```
##
##  Overdispersion test
##
## data:  poisson_model
## z = 20.927, p-value < 2.2e-16
## alternative hypothesis: true dispersion is greater than 1
## sample estimates:
## dispersion
##   4.560182
```

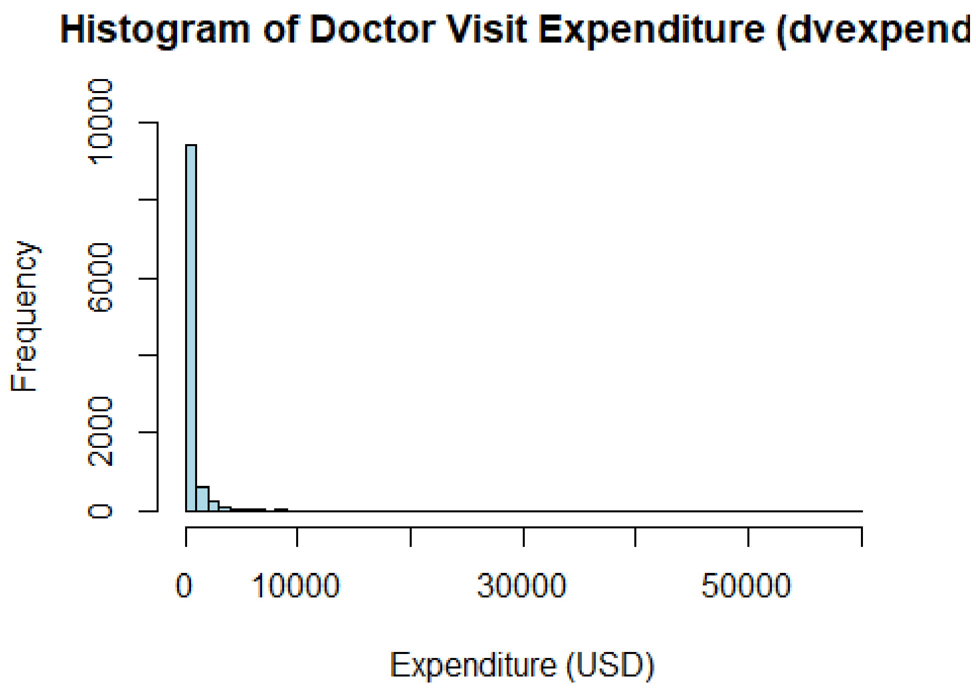
```
# Negative Binomial model
nb_model <- glm.nb(
  dvisit ~ general + mental + bmi + log_income + age + gender +
    ethnicity + education + region + hypertension + hyperlipidemia,
  data = meps
)

summary(nb_model)
```

```
##
## Call:
## glm.nb(formula = dvisit ~ general + mental + bmi + log_income +
##     age + gender + ethnicity + education + region + hypertension +
##     hyperlipidemia, data = meps, init.theta = 0.6451146395, link = log)
##
## Coefficients:
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)     -1.534158   0.129581 -11.839  < 2e-16 ***
## general2         0.292543   0.047022   6.221 4.93e-10 ***
## general3         0.532107   0.051511  10.330  < 2e-16 ***
## general4         0.837430   0.065882  12.711  < 2e-16 ***
## general5         1.243109   0.097786  12.713  < 2e-16 ***
## mental2          0.032204   0.043054   0.748 0.454470
## mental3          0.044169   0.047421   0.931 0.351644
## mental4          0.387132   0.070476   5.493 3.95e-08 ***
## mental5          0.636142   0.123208   5.163 2.43e-07 ***
## bmi              0.008451   0.002355   3.588 0.000333 ***
## log_income       0.010658   0.007575   1.407 0.159407
## age              0.012860   0.001233  10.431  < 2e-16 ***
## genderMale      -0.676785   0.029526 -22.922  < 2e-16 ***
## ethnicity2      -0.203185   0.038315  -5.303 1.14e-07 ***
## ethnicity3      -0.148424   0.162606  -0.913 0.361357
## ethnicity4      -0.168821   0.054259  -3.111 0.001862 **
## education        0.085426   0.005407  15.800  < 2e-16 ***
## region2          0.010259   0.048498   0.212 0.832477
## region3         -0.211824   0.043497  -4.870 1.12e-06 ***
## region4         -0.164141   0.046660  -3.518 0.000435 ***
## hypertension     0.309565   0.038051   8.136 4.10e-16 ***
## hyperlipidemia   0.387506   0.037939  10.214  < 2e-16 ***
## ---
```

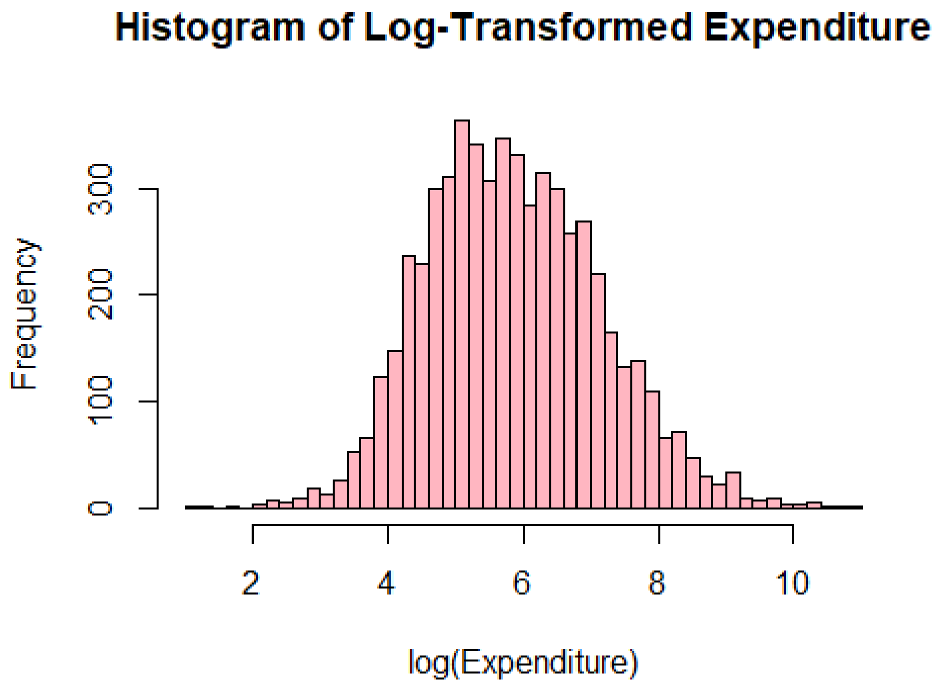
```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(0.6451) family taken to be
1)
##
##     Null deviance: 13099  on 10637  degrees of freedom
## Residual deviance: 10411  on 10616  degrees of freedom
## AIC: 37796
##
## Number of Fisher Scoring iterations: 1
##
##
##               Theta:  0.6451
##           Std. Err.:  0.0148
##
##  2 x log-likelihood:  -37749.8480

AIC(poisson_model, nb_model)   # Model comparison

##               df      AIC
## poisson_model 22 51956.65
## nb_model      23 37795.85

# TWO-PART MODEL FOR EXPENDITURE (dvexpend)

# Part 1: Logistic regression (any expenditure)
logit_model <- glm(
  any_dv ~ general + mental + bmi + log_income + age + gender +
    ethnicity + education + region + hypertension + hyperlipidemia,
  family = binomial(link = "logit"),
  data = meps
)

# Part 2: Gamma model among positive expenditure users
gamma_model <- glm(
  dvexpend ~ general + mental + bmi + log_income + age + gender +
    ethnicity + education + region + hypertension + hyperlipidemia,
  family = Gamma(link = "log"),
  data = subset(meps, dvexpend > 0)
)

# MODEL RESULTS TABLES (for report)

logit_OR <- tidy(logit_model, conf.int = TRUE, exponentiate = TRUE)
gamma_EXP <- tidy(gamma_model, conf.int = TRUE, exponentiate = TRUE)
nb_IRR <- tidy(nb_model, conf.int = TRUE, exponentiate = TRUE)

results_logit_subset <- logit_OR %>%
  filter(term %in% c("general5", "mental5", "log_income", "age",
                     "genderMale", "hypertension", "hyperlipidemia"))

results_gamma_subset <- gamma_EXP %>%
  filter(term %in% c("general5", "age", "genderMale", "hypertension"))

results_nb_subset <- nb_IRR %>%
```

```
  filter(term %in% c("general5", "mental5", "age", "genderMale", "hyperten
sion"))

results_logit_subset

## # A tibble: 7 × 7
##   term           estimate std.error statistic  p.value conf.low conf.hi
gh
##   <chr>             <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <db
l>
## 1 general5          3.92    0.196        6.96 3.41e-12    2.69      5.8
2
## 2 mental5           1.40    0.237        1.41 1.60e- 1    0.884     2.2
4
## 3 log_income        1.06    0.0113       5.31 1.09e- 7    1.04      1.0
9
## 4 age               1.02    0.00179      9.37 7.31e-21    1.01      1.0
2
## 5 genderMale        0.397   0.0438     -21.1  3.54e-99    0.364     0.4
32
## 6 hypertension      1.81    0.0602       9.88 5.20e-23    1.61      2.0
4
## 7 hyperlipidemia    2.21    0.0621      12.7  3.36e-37    1.95      2.4
9

results_gamma_subset

## # A tibble: 4 × 7
##   term         estimate std.error statistic  p.value conf.low conf.high
##   <chr>           <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
## 1 general5        3.13    0.168        6.80 1.19e-11    2.27      4.38
## 2 age             1.01    0.00232      4.25 2.21e- 5    1.01      1.01
## 3 genderMale      0.804   0.0561      -3.89 9.97e- 5    0.720     0.899
## 4 hypertension    1.13    0.0683       1.85 6.46e- 2    0.992     1.30

results_nb_subset

## # A tibble: 5 × 7
##   term         estimate std.error statistic   p.value conf.low conf.hig
h
##   <chr>           <dbl>     <dbl>     <dbl>     <dbl>    <dbl>     <dbl
>
## 1 general5        3.47    0.0978      12.7  5.03e- 37    2.88      4.19
## 2 mental5         1.89    0.123        5.16 2.43e-  7    1.49      2.42
## 3 age             1.01    0.00123     10.4  1.79e- 25    1.01      1.02
## 4 genderMale      0.508   0.0295     -22.9  2.81e-116    0.480     0.53
9
## 5 hypertension    1.36    0.0381       8.14 4.10e- 16    1.27      1.47

# FIGURES

# FIGURE 1: Histogram of expenditure
hist(meps$dvexpend,
     main = "Histogram of Doctor Visit Expenditure (dvexpend)",
     xlab = "Expenditure (USD)",
```

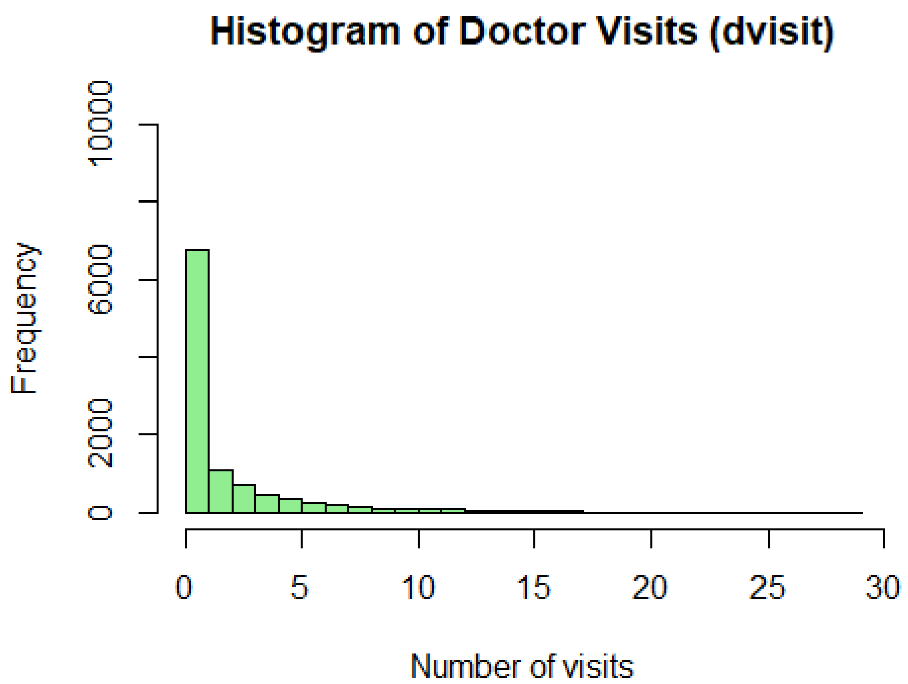
```
      col = "lightblue", border = "black",
      breaks = 60, ylim = c(0, 10000))
```



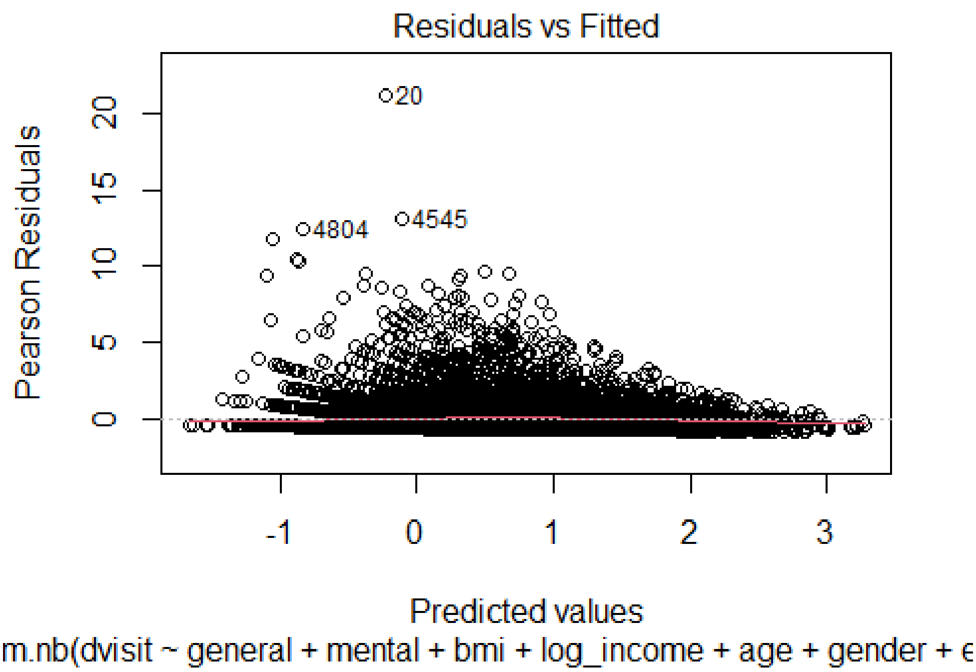

```
# FIGURE 2: Log-transformed expenditure
hist(log(meps$dvexpend[meps$dvexpend > 0]),
      main = "Histogram of Log-Transformed Expenditure",
      xlab = "log(Expenditure)",
      col = "lightpink", border = "black",
      breaks = 40)
```



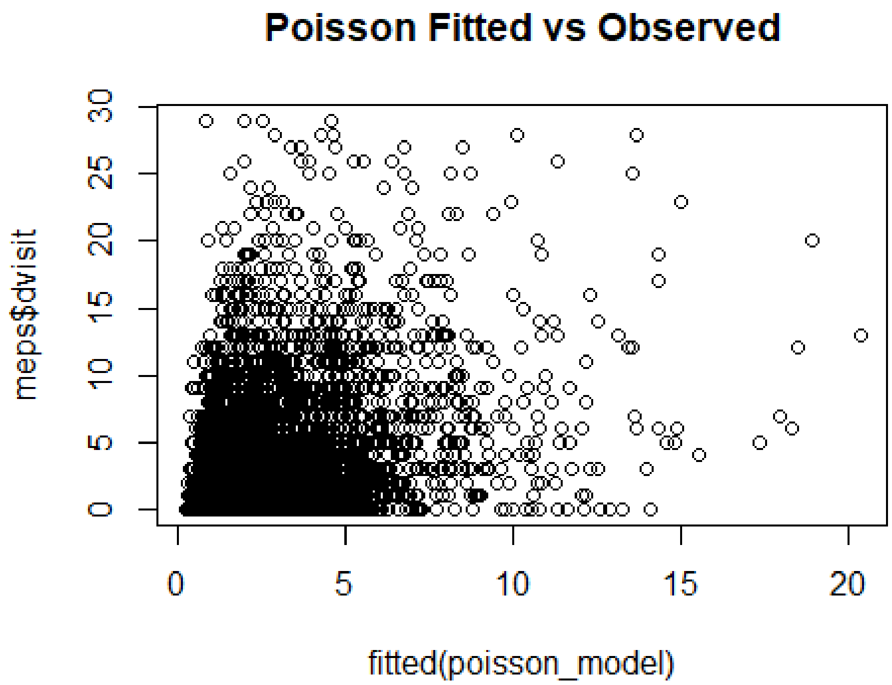

```r
# FIGURE 3: Histogram of doctor visits
hist(meps$dvisit,
     main = "Histogram of Doctor Visits (dvisit)",
     xlab = "Number of visits",
     col = "lightgreen", border = "black",
     breaks = 30, ylim = c(0, 10000))
```



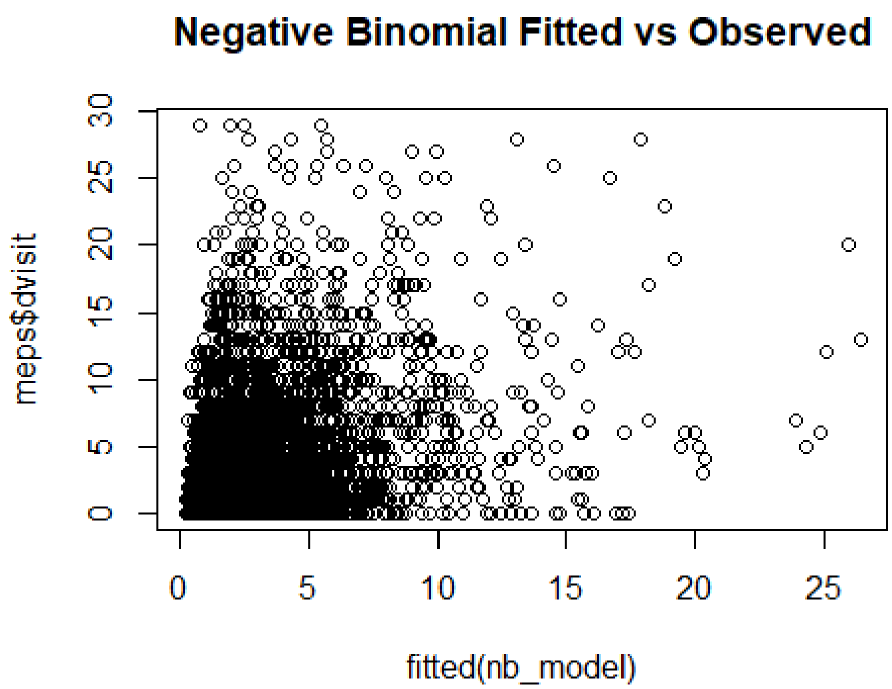
```r
# FIGURE 4a-4c: Additional diagnostic plots
plot(nb_model, which = 1)
```



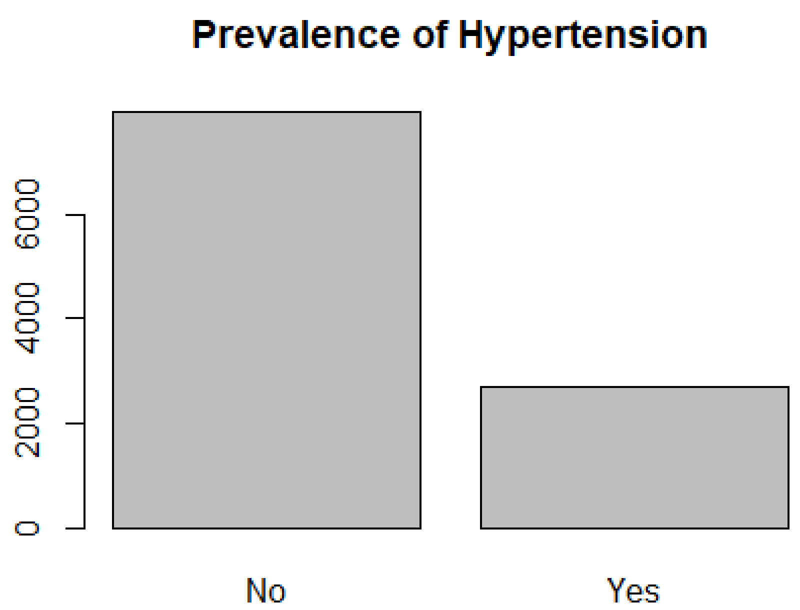

```r
plot(fitted(poisson_model), meps$dvisit,
     main = "Poisson Fitted vs Observed")
```




```r
plot(fitted(nb_model), meps$dvisit,
     main = "Negative Binomial Fitted vs Observed")
```




```r
barplot(table(meps$hypertension),
        main = "Prevalence of Hypertension",
        names.arg = c("No", "Yes"), col = "grey")
```

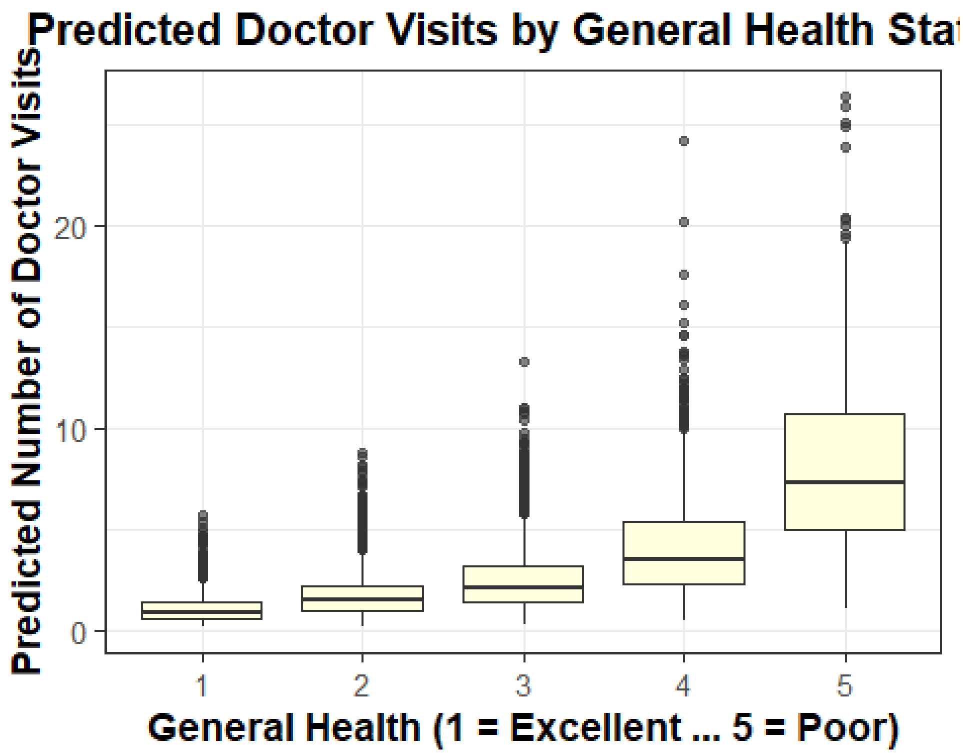



```r
# FIGURE 5: Predicted utilisation by general health (boxplot)
# Ensure general is ordered for plotting (factor already created above)
meps$general <- factor(meps$general,
```

```
                        levels = sort(levels(meps$general)),
                        ordered = TRUE)

meps$pred_nb <- predict(nb_model, type = "response")

ggplot(meps, aes(x = general, y = pred_nb)) +
  geom_boxplot(fill = "lightyellow", outlier.size = 1.5, outlier.alpha = 0
.6) +
  labs(
    title = "Predicted Doctor Visits by General Health Status",
    x = "General Health (1 = Excellent ... 5 = Poor)",
    y = "Predicted Number of Doctor Visits"
  ) +
  theme_bw(base_size = 14) +
  theme(
    plot.title  = element_text(face = "bold", hjust = 0.5),
    axis.title  = element_text(face = "bold")
  )
```



R Code

```
---
title: "Full R code"
output: word_document
---

```{r}
# Load required packages
library(MASS)
library(AER)
library(broom)
library(dplyr)
library(ggplot2)


meps <- read.table("meps.txt", header = TRUE)

# Convert categorical variables and create derived variables
meps$general    <- factor(meps$general)
meps$mental     <- factor(meps$mental)
meps$ethnicity  <- factor(meps$ethnicity)
meps$region     <- factor(meps$region)
meps$gender     <- factor(meps$gender, levels = c(0,1), labels =
c("Female","Male"))
meps$log_income <- log(meps$income + 1)
meps$any_dv     <- as.numeric(meps$dvexpend > 0)   # Indicator for
positive expenditure


#TABLE 1: Demographic and health characteristics of the analysis sample
library(dplyr)

demographic_table <- meps %>%
  summarise(
    Mean_Age = mean(age, na.rm = TRUE),
    SD_Age = sd(age, na.rm = TRUE),
    Female = mean(gender == "Female") * 100,
    Male = mean(gender == "Male") * 100,
    Poor_General_Health = mean(general == 5) * 100,
    Poor_Mental_Health = mean(mental == 5) * 100,
    Hypertension = mean(hypertension == 1) * 100,
    Hyperlipidemia = mean(hyperlipidemia == 1) * 100,
    Mean_BMI = mean(bmi, na.rm = TRUE),
    SD_BMI = sd(bmi, na.rm = TRUE),
    Mean_Log_Income = mean(log_income, na.rm = TRUE),
    SD_Log_Income = sd(log_income, na.rm = TRUE)
  )

demographic_table

# TABLE 2: Summary statistics for dvexpend

n <- nrow(meps)
mean_dvexpend   <- mean(meps$dvexpend, na.rm = TRUE)
median_dvexpend <- median(meps$dvexpend, na.rm = TRUE)
sd_dvexpend     <- sd(meps$dvexpend, na.rm = TRUE)
min_dvexpend    <- min(meps$dvexpend, na.rm = TRUE)
```
```

```
max_dvexpend    <- max(meps$dvexpend, na.rm = TRUE)
zero_share_dvexpend <- mean(meps$dvexpend == 0)

cat("Summary of dvexpend:\n",
    "N =", n, "\n",
    "Mean =", mean_dvexpend, "\n",
    "Median =", median_dvexpend, "\n",
    "Standard deviation =", sd_dvexpend, "\n",
    "Min =", min_dvexpend, "\n",
    "Max =", max_dvexpend, "\n",
    "Share zero =", zero_share_dvexpend, "\n\n")



# TABLE 3: Summary statistics for dvisit


mean_dvisit   <- mean(meps$dvisit, na.rm = TRUE)
median_dvisit <- median(meps$dvisit, na.rm = TRUE)
var_dvisit    <- var(meps$dvisit, na.rm = TRUE)
min_dvisit    <- min(meps$dvisit, na.rm = TRUE)
max_dvisit    <- max(meps$dvisit, na.rm = TRUE)
zero_share_dvisit <- mean(meps$dvisit == 0)

summary_dvisit <- data.frame(
  Statistic = c("N", "Mean", "Median", "Variance", "Minimum",
"Maximum", "Share of zeros"),
  Value     = c(n, mean_dvisit, median_dvisit, var_dvisit,
                min_dvisit, max_dvisit, zero_share_dvisit)
)

cat("Summary of dvisit:\n",
    "N =", n, "\n",
    "Mean =", mean_dvisit, "\n",
    "Median =", median_dvisit, "\n",
    "Variance =", var_dvisit, "\n",
    "Min =", min_dvisit, "\n",
    "Max =", max_dvisit, "\n",
    "Share zero =", zero_share_dvisit, "\n\n")

summary_dvisit


# MODEL SELECTION FOR UTILISATION (dvisit)

# Poisson model
poisson_model <- glm(
  dvisit ~ general + mental + bmi + log_income + age + gender +
    ethnicity + education + region + hypertension + hyperlipidemia,
  family = poisson(link = "log"),
  data = meps
)

summary(poisson_model)
dispersiontest(poisson_model)   # Overdispersion test

# Negative Binomial model
nb_model <- glm.nb(
```

```
  dvisit ~ general + mental + bmi + log_income + age + gender +
    ethnicity + education + region + hypertension + hyperlipidemia,
  data = meps
)

summary(nb_model)
AIC(poisson_model, nb_model)   # Model comparison


# TWO-PART MODEL FOR EXPENDITURE (dvexpend)

# Part 1: Logistic regression (any expenditure)
logit_model <- glm(
  any_dv ~ general + mental + bmi + log_income + age + gender +
    ethnicity + education + region + hypertension + hyperlipidemia,
  family = binomial(link = "logit"),
  data = meps
)

# Part 2: Gamma model among positive expenditure users
gamma_model <- glm(
  dvexpend ~ general + mental + bmi + log_income + age + gender +
    ethnicity + education + region + hypertension + hyperlipidemia,
  family = Gamma(link = "log"),
  data = subset(meps, dvexpend > 0)
)

# MODEL RESULTS TABLES (for report)

logit_OR <- tidy(logit_model, conf.int = TRUE, exponentiate = TRUE)
gamma_EXP <- tidy(gamma_model, conf.int = TRUE, exponentiate = TRUE)
nb_IRR <- tidy(nb_model, conf.int = TRUE, exponentiate = TRUE)

results_logit_subset <- logit_OR %>%
  filter(term %in% c("general5", "mental5", "log_income", "age",
                     "genderMale", "hypertension", "hyperlipidemia"))

results_gamma_subset <- gamma_EXP %>%
  filter(term %in% c("general5", "age", "genderMale", "hypertension"))

results_nb_subset <- nb_IRR %>%
  filter(term %in% c("general5", "mental5", "age", "genderMale",
"hypertension"))

results_logit_subset
results_gamma_subset
results_nb_subset


# FIGURES

# FIGURE 1: Histogram of expenditure
hist(meps$dvexpend,
     main = "Histogram of Doctor Visit Expenditure (dvexpend)",
     xlab = "Expenditure (USD)",
     col = "lightblue", border = "black",
     breaks = 60, ylim = c(0, 10000))
```

```
# FIGURE 2: Log-transformed expenditure
hist(log(meps$dvexpend[meps$dvexpend > 0]),
     main = "Histogram of Log-Transformed Expenditure",
     xlab = "log(Expenditure)",
     col = "lightpink", border = "black",
     breaks = 40)

# FIGURE 3: Histogram of doctor visits
hist(meps$dvisit,
     main = "Histogram of Doctor Visits (dvisit)",
     xlab = "Number of visits",
     col = "lightgreen", border = "black",
     breaks = 30, ylim = c(0, 10000))

# FIGURE 4a-4c: Additional diagnostic plots
plot(nb_model, which = 1)
plot(fitted(poisson_model), meps$dvisit,
     main = "Poisson Fitted vs Observed")
plot(fitted(nb_model), meps$dvisit,
     main = "Negative Binomial Fitted vs Observed")
barplot(table(meps$hypertension),
        main = "Prevalence of Hypertension",
        names.arg = c("No", "Yes"), col = "grey")

# FIGURE 5: Predicted utilisation by general health (boxplot)
# Ensure general is ordered for plotting (factor already created above)
meps$general <- factor(meps$general,
                       levels = sort(levels(meps$general)),
                       ordered = TRUE)

meps$pred_nb <- predict(nb_model, type = "response")

ggplot(meps, aes(x = general, y = pred_nb)) +
  geom_boxplot(fill = "lightyellow", outlier.size = 1.5, outlier.alpha
= 0.6) +
  labs(
    title = "Predicted Doctor Visits by General Health Status",
    x = "General Health (1 = Excellent ... 5 = Poor)",
    y = "Predicted Number of Doctor Visits"
  ) +
  theme_bw(base_size = 14) +
  theme(
    plot.title  = element_text(face = "bold", hjust = 0.5),
    axis.title  = element_text(face = "bold")
  )
```